Jinyao DeSandies

Professor Marisa Cameron

Economics of Crime

Project 4 Write-Up

The state of gun related incidents in the United States is concerning. Over the period of study for this project, it was on the rise, in part as gun ownership itself rose in most of the United States (Gallup 2019). Three visualizations, a hex map, a heat map, and a bump chart will show how different states experienced varying level of gun violence, from 2014 to 2017. These trends match with certain economic hypothesis, which suggest that better quality of life and stricter gun control may be the optimal solutions to this national concern.

The first visualization is a hex map describing state-specific gun violence impact rate. The use of a hex map rather than a realistic map was important to address the Alaska effect, where some states large in area potentially disproportionally, visually overshadow other states (Taylor 2017). The data was also rated to a per 100,000 bases, allowing an accurate rendition of gun violence in proportion to the population of the state. It is notable that violent crime increased in most states over the time span, but decreased in Vermont, where gun control legislations were in progress (State of Vermont 2018). This alludes to an effect Duggan coined as “more guns more crime” in 2001, arguing ease of obtaining guns increase criminal gun ownership not only because criminals will have more access to firearms, but feel a necessity for them because victims are more likely to be armed. Similarly, Canada did not experience a drastic rise in crime, as more strict gun control and even freezes kept gun ownership to below 10%, far under the United States. The low gun ownership rate corresponded to low rates of gun-related violence, unlike in the US (Lukiv 2026).

The next visualization on the dashboard was a crime heat map, which showed total rated gun violence impact by month and day of the week, in a selected state. The darker the color, the more gun-related incidents occur. For Florida, gun related incidents were most common on Sundays during the summer. Increased crime during the summer weekends can be explained by an environmental factor: it is statistically proven increased aggressiveness during high levels of heat (Heilmann, Kahn, and Tang 2021). The trend can also be analyzed from the perspective of gun-related incident type, which was recorded by the Gun Violence Archive , according to their methodology note in 2012. Some incidence types are more common during the hot, tourist-filled Florida summer, such as accidental injury: by children, when school is not in session, or by adults, when hunting trips are organized. Some incidents are more likely to occur on the weekends: familicides or drivebys, as individuals are often in more informal or potentially predictable locations. The trend thus may explain the trends in Florida's crime distribution.

Finally, a bump chart in the bottom of the dashboard describes the movement in state rankings of gun-related incidents. There are shifting in rankings over time, but some states generally stay in the top 3. These include Louisiana, Illinois, and Delaware. Florida never broke top 10, in part because it does not fit the architype of a high gun violence state. States with significantly high gun violence tended to either be dominated by urban areas (Illinois, Delaware, Maryland) or a high level of poverty (US South, Midwest). This aligns with the supply and demand model of crime proposed by Cook in 1986: high density in urban areas create more opportunity, or supply for crime, whereas poverty drives more individuals to feel a need to commit crime to make ends meet, which can be encapsulated as demand. Florida, with its large elderly and affluent suburban population, does not experience these drivers, and thus experience non-exceptional levels of gun violence (US Census 2024).

In conclusion, this series of visualizations effectively show variation in gun violence across America, over time. Gun violence increased in most states but decreased in states where gun ownership did not increase. Month and day of the week also mattered: Gun violence in Florida was especially frequent on weekends and during the summer, in part due to lifestyle patterns and the phycological brutalizing effect of temperature. Lastly, as the supply and demand model of crime suggested, more urbanized or poorer states experienced more gun violence. Thus, a general increase in standard of living or gun control emerges as potential means to addressing the phenomenon.

Appendix

Bibliography

Cook, PJ. 1987. "The Demand and Supply of Criminal Opportunities." *Jstor.org*. https://doi.org/10.2307/1147515.

Duggan, Mark. 2001. "More Guns, More Crime." *Journal of Political Economy* 109 (5): 1086–1114. https://doi.org/10.1086/322833.

Gallup. 2019. "Guns." Gallup.com. April 2019. https://news.gallup.com/poll/1645/guns.aspx.

Gun Violence Archive. 2012. "General Methodology | Gun Violence Archive." Gunviolencearchive.org. 2012. https://www.gunviolencearchive.org/methodology.

Heilmann, Kilian, Matthew E. Kahn, and Cheng Keat Tang. 2021. "The Urban Crime and Heat Gradient in High and Low Poverty Areas." *Journal of Public Economics* 197 (May): 104408. https://doi.org/10.1016/j.jpubeco.2021.104408.

Jaroslav Lukiv. 2026. "What Are Canada's Gun Ownership Laws?" *BBC*, February 11, 2026. https://www.bbc.com/news/articles/cwy8eyz4ee0o.

State of Vermont. 2018. "New Vermont Gun Laws FAQs | Department of Public Safety." Vermont.gov. 2018. https://dps.vermont.gov/content/new-vermont-gun-laws-faqs.

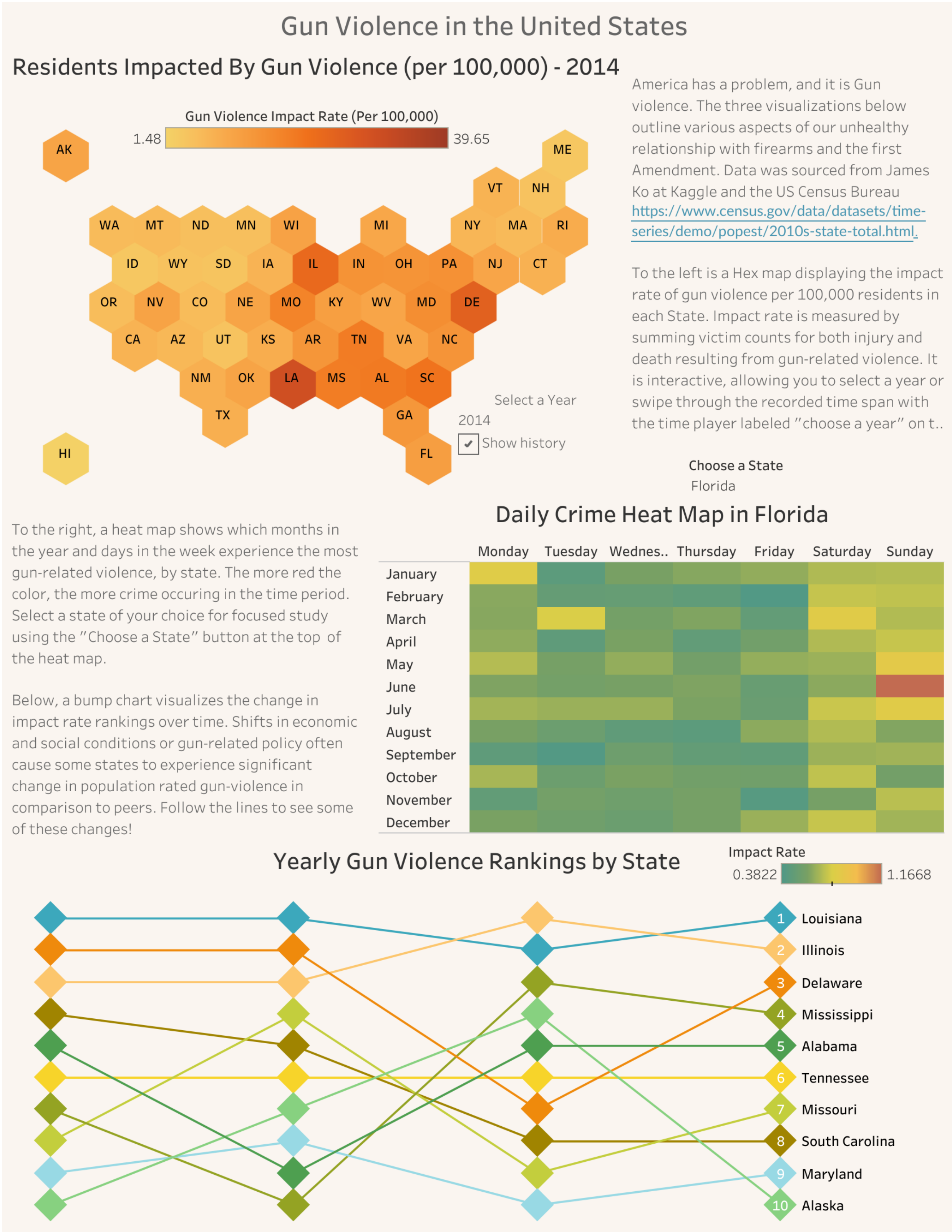
Taylor, Kevin. 2017. "How to Use Hex-Tile Maps to Eliminate the Alaska Effect." Tableau. 2017. https://www.tableau.com/blog/viz-whiz-hex-tile-maps-64713.

US Census. 2024. "QuickFacts: Florida." Census Bureau QuickFacts. United States Census Bureau. 2024. https://www.census.gov/quickfacts/fact/table/FL/PST045224.

Author Statement:

No AI was used in the process of creating this project.

Dashboard:



Gun Violence in the United States
Residents Impacted By Gun Violence (per 100,000) - 2014
Gun Violence Impact Rate (Per 100,000)
1.48
39.65
AK
ME
VT
NH
WA
MT
ND
MN
WI
MI
NY
MA
RI
ID
WY
SD
IA
IL
IN
OH
PA
NJ
CT
OR
NV
CO
NE
MO
KY
WV
MD
DE
CA
AZ
UT
KS
AR
TN
VA
NC
NM
OK
LA
MS
AL
SC
TX
GA
FL
HI
Select a Year
2014
Show history
America has a problem, and it is Gun violence. The three visualizations below outline various aspects of our unhealthy relationship with firearms and the first Amendment. Data was sourced from James Ko at Kaggle and the US Census Bureau https://www.census.gov/data/datasets/time-series/demo/popest/2010s-state-total.html.
To the left is a Hex map displaying the impact rate of gun violence per 100,000 residents in each State. Impact rate is measured by summing victim counts for both injury and death resulting from gun-related violence. It is interactive, allowing you to select a year or swipe through the recorded time span with the time player labeled "choose a year" on t..
Choose a State
Florida
Daily Crime Heat Map in Florida
To the right, a heat map shows which months in the year and days in the week experience the most gun-related violence, by state. The more red the color, the more crime occuring in the time period. Select a state of your choice for focused study using the "Choose a State" button at the top of the heat map.
Below, a bump chart visualizes the change in impact rate rankings over time. Shifts in economic and social conditions or gun-related policy often cause some states to experience significant change in population rated gun-violence in comparison to peers. Follow the lines to see some of these changes!
Monday
Tuesday
Wednes..
Thursday
Friday
Saturday
Sunday
January
February
March
April
May
June
July
August
September
October
November
December
Yearly Gun Violence Rankings by State
Impact Rate
0.3822
1.1668
1 Louisiana
2 Illinois
3 Delaware
4 Mississippi
5 Alabama
6 Tennessee
7 Missouri
8 South Carolina
9 Maryland
10 Alaska